

Lecture 4: Theoretical Fundamentals of Dynamic Programming

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1 Overview

Last lecture: MDP, DP, Value Iteration (VI), Policy Iteration (PI).

This lecture:

- Deepen the mathematical formalism behind the MDP framework.
- Revisit the Bellman equations and introduce their corresponding operators.
- Re-visit the paradigm of dynamic programming: VI and PI.

Next lectures: Approximate, sampled versions of these paradigms, mainly in the absence of perfect knowledge of the environment.

2 Preliminaries

2.1 Normed Vector Spaces

Definition 1 (Normed Vector Space). *A normed vector space is a vector space $\mathcal{X}$ equipped with a norm $\|\cdot\|$ on the elements of $\mathcal{X}$. A norm is a mapping $\mathcal{X} \rightarrow \mathbb{R}$ such that:*

1. $\|x\| \geq 0, \forall x \in \mathcal{X}$ and if $\|x\| = 0$ then $x = 0$.
2. $\|\alpha x\| = |\alpha| \|x\|$ (homogeneity).
3. $\|x_1 + x_2\| \leq \|x_1\| + \|x_2\|$ (triangle inequality).

For this lecture, we consider:

- Vector spaces: $\mathcal{X} = \mathbb{R}^d$.
- Norms:
 - Max-norm / L_∞ norm: $\|\cdot\|_\infty$.
 - (Weighted) L_2 norms: $\|\cdot\|_{2,p}$.

2.2 Contraction Mapping

Definition 2 (Contraction Mapping). *Let $\mathcal{X}$ be a vector space equipped with a norm $\|\cdot\|$. A mapping $\mathcal{T} : \mathcal{X} \rightarrow \mathcal{X}$ is an α -contraction mapping if for any $x_1, x_2 \in \mathcal{X}$, there exists $\alpha \in [0, 1)$ such that:*

$$\|\mathcal{T}x_1 - \mathcal{T}x_2\| \leq \alpha \|x_1 - x_2\| \quad (1)$$

If $\alpha \in [0, 1]$, then we call $\mathcal{T}$ non-expanding.

Remark 1. Every contraction is also (by definition) Lipschitz, thus it is also continuous. In particular, this means: If $x_n \rightarrow_{\|\cdot\|} x$, then $\mathcal{T}x_n \rightarrow_{\|\cdot\|} \mathcal{T}x$.

2.3 Fixed Point

Definition 3 (Fixed Point). *A point/vector $x \in \mathcal{X}$ is a fixed point of an operator $\mathcal{T}$ if $\mathcal{T}x = x$.*

2.4 Banach Fixed Point Theorem

Theorem 1 (Banach Fixed Point Theorem). *Let $\mathcal{X}$ be a complete normed vector space equipped with a norm $\|\cdot\|$ and $\mathcal{T} : \mathcal{X} \rightarrow \mathcal{X}$ be a γ -contraction mapping, then:*

1. $\mathcal{T}$ has a unique fixed point $x^* \in \mathcal{X}$: $\exists! x^* \in \mathcal{X}$ such that $\mathcal{T}x^* = x^*$.
2. For all $x_0 \in \mathcal{X}$, the sequence $x_{n+1} = \mathcal{T}x_n$ converges to x^* in a geometric fashion:

$$\|x_n - x^*\| \leq \gamma^n \|x_0 - x^*\| \quad (2)$$

Thus $\lim_{n \rightarrow \infty} \|x_n - x^*\| \leq \lim_{n \rightarrow \infty} (\gamma^n \|x_0 - x^*\|) = 0$.

3 Markov Decision Processes and Dynamic Programming (Recap)

3.1 MDP Definition

Markov Decision Processes (MDPs) formally describe an environment:

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle \quad (3)$$

Almost all RL problems can be formalised as MDPs (e.g., Optimal control, Bandits, Partially observable problems).

3.2 Value Functions

Definition 4 (State Value Function). *For a policy π :*

$$v_\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid s_0 = s; \pi \right] \quad (4)$$

Definition 5 (Action Value Function). *For a policy π :*

$$q_\pi(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid s_0 = s, a_0 = a; \pi \right] \quad (5)$$

Optimal value functions: $q^* = \max_\pi q_\pi$ and $v^* = \max_\pi v_\pi$.

3.3 Bellman Equations

Theorem 2 (Bellman Expectation Equations). *Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, for any policy π , the value functions obey the following expectation equations:*

$$v_\pi(s) = \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) v_\pi(s') \right] \quad (1)$$

$$q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|a, s) \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s', a') \quad (2)$$

Theorem 3 (Bellman Optimality Equations). *Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, the optimal value functions obey the following expectation equations:*

$$v^*(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) v^*(s') \right] \quad (3)$$

$$q^*(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|a, s) \max_{a' \in \mathcal{A}} q^*(s', a') \quad (4)$$

4 Bellman Operators

4.1 The Bellman Optimality Operator T^*

Definition 6 (Bellman Optimality Operator T_V^*). Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, let $\mathcal{V} \equiv \mathcal{V}_{\mathcal{S}}$ be the space of bounded real-valued functions over $\mathcal{S}$. We define, point-wise, the Bellman Optimality operator $T_V^* : \mathcal{V} \rightarrow \mathcal{V}$ as:

$$(T_V^* f)(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) f(s') \right], \quad \forall f \in \mathcal{V} \quad (5)$$

As a common convention we drop the index $\mathcal{V}$ and simply use $T^* = T_V^*$.

4.1.1 Properties of the Bellman Operator T^*

1. It has one unique fixed point v^* : $T^* v^* = v^*$.
2. T^* is a γ -contraction wrt. to $\|\cdot\|_{\infty}$:

$$\|T^* v - T^* u\|_{\infty} \leq \gamma \|v - u\|_{\infty}, \quad \forall u, v \in \mathcal{V} \quad (6)$$

3. T^* is monotonic:

$$\forall u, v \in \mathcal{V} \text{ s.t. } u \leq v \text{ component-wise, then } T^* u \leq T^* v \quad (7)$$

4.1.2 Proofs of Properties

Proposition 1. T^* is a γ -contraction wrt. to $\|\cdot\|_{\infty}$.

Proof.

$$|T^* v(s) - T^* u(s)| = \left| \max_a [r(s, a) + \gamma \mathbb{E}_{s'|s, a} v(s')] - \max_b [r(s, b) + \gamma \mathbb{E}_{s''|s, b} u(s'')] \right| \quad (6)$$

$$\leq \max_a \left| [r(s, a) + \gamma \mathbb{E}_{s'|s, a} v(s')] - [r(s, a) + \gamma \mathbb{E}_{s'|s, a} u(s')] \right| \quad (7)$$

$$= \gamma \max_a \left| \mathbb{E}_{s'|s, a} [v(s') - u(s')] \right| \quad (8)$$

$$\leq \gamma \max_{s'} |v(s') - u(s')| \quad (9)$$

Thus we get:

$$\|T^* v - T^* u\|_{\infty} \leq \gamma \|v - u\|_{\infty}, \quad \forall u, v \in \mathcal{V} \quad (8)$$

Note: Step (6)-(7) uses: $|\max_a f(a) - \max_b g(b)| \leq \max_a |f(a) - g(a)|$. $\square$

Proposition 2. T^* is monotonic.

Proof. Given $v(s) \leq u(s), \forall s \Rightarrow r(s, a) + \mathbb{E}_{s'|s, a} v(s') \leq r(s, a) + \mathbb{E}_{s'|s, a} u(s')$.

$$T^* v(s) - T^* u(s) = \max_a [r(s, a) + \gamma \mathbb{E}_{s'|s, a} v(s')] - \max_b [r(s, b) + \gamma \mathbb{E}_{s''|s, b} u(s'')] \quad (10)$$

$$\leq \max_a ([r(s, a) + \gamma \mathbb{E}_{s'|s, a} v(s')] - [r(s, a) + \gamma \mathbb{E}_{s'|s, a} u(s')]) \quad (11)$$

$$\leq 0, \quad \forall s. \quad (12)$$

Thus $T^* v(s) \leq T^* u(s), \forall s \in \mathcal{S}$. $\square$

4.2 Value Iteration through the lens of the Bellman Operator

Value Iteration

1. Start with v_0 .
2. Update values: $v_{k+1} = T^* v_k$.
3. As $k \rightarrow \infty, v_k \rightarrow_{\|\cdot\|_{\infty}} v^*$.

Proof of Convergence. Direct application of the Banach Fixed Point Theorem.

$$\|v_k - v^*\|_\infty = \|T^* v_{k-1} - v^*\|_\infty \quad (9)$$

$$= \|T^* v_{k-1} - T^* v^*\|_\infty \quad (\text{fixed point prop.}) \quad (10)$$

$$\leq \gamma \|v_{k-1} - v^*\|_\infty \quad (\text{contraction prop.}) \quad (11)$$

$$\leq \gamma^k \|v_0 - v^*\|_\infty \quad (\text{iterative application}) \quad (12)$$

□

4.3 The Bellman Expectation Operator T^π

Definition 7 (Bellman Expectation Operator). Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, let $\mathcal{V} \equiv \mathcal{V}_{\mathcal{S}}$ be the space of bounded real-valued functions over $\mathcal{S}$. For any policy $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, we define, point-wise, the Bellman Expectation operator $T_V^\pi : \mathcal{V} \rightarrow \mathcal{V}$ as:

$$(T_V^\pi f)(s) = \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) f(s') \right], \quad \forall f \in \mathcal{V} \quad (13)$$

4.3.1 Properties of the Bellman Operator T^π

1. It has one unique fixed point v_π : $T^\pi v_\pi = v_\pi$.
2. T^π is a γ -contraction wrt. to $\|\cdot\|_\infty$:

$$\|T^\pi v - T^\pi u\|_\infty \leq \gamma \|v - u\|_\infty, \quad \forall u, v \in \mathcal{V} \quad (13)$$

3. T^π is monotonic:

$$\forall u, v \in \mathcal{V} \text{ s.t. } u \leq v \text{ component-wise, then } T^\pi u \leq T^\pi v \quad (14)$$

4.3.2 Proofs of Properties

Proposition 3. T^π is a γ -contraction wrt. to $\|\cdot\|_\infty$.

Proof.

$$T^\pi v(s) - T^\pi u(s) = \sum_a \pi(a|s) [r(s, a) + \gamma \mathbb{E}_{s'|s, a} v(s') - r(s, a) - \gamma \mathbb{E}_{s'|s, a} u(s')] \quad (15)$$

$$= \gamma \sum_a \pi(a|s) \mathbb{E}_{s'|s, a} [v(s') - u(s')] \quad (16)$$

$$\Rightarrow |T^\pi v(s) - T^\pi u(s)| \leq \gamma \max_{s'} |v(s') - u(s')| \quad (14)$$

Thus we get:

$$\|T^\pi v - T^\pi u\|_\infty \leq \gamma \|v - u\|_\infty, \quad \forall u, v \in \mathcal{V} \quad (17)$$

Note: (14) gives us also Prop. (3), monotonicity of T^π . □

4.4 Policy Evaluation

(Iterative) Policy Evaluation

1. Start with v_0 .
2. Update values: $v_{k+1} = T^\pi v_k$.
3. As $k \rightarrow \infty$, $v_k \rightarrow_{\|\cdot\|_\infty} v_\pi$.

Proof: Direct application of the Banach Fixed Point Theorem.

4.5 Summary: Dynamic Programming with Bellman Operators

Value Iteration

- Start with v_0 .
- Update values: $v_{k+1} = T^*v_k$.

Policy Iteration

- Start with π_0 .
- Iterate:
 1. **Policy Evaluation:** v_{π_i}
(E.g. For instance, by iterating $T^\pi : v_k = T^{\pi_i}v_{k-1} \Rightarrow v_k \rightarrow v^{\pi_i}$ as $k \rightarrow \infty$)
 2. **Greedy Improvement:** $\pi_{i+1} = \arg \max_a q_{\pi_i}(s, a)$

4.6 Bellman Operators for Q -functions

Definition 8 (Bellman Expectation Operator for Q). Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, let $\mathcal{Q} \equiv \mathcal{Q}_{\mathcal{S}, \mathcal{A}}$ be the space of bounded real-valued functions over $\mathcal{S} \times \mathcal{A}$. For any policy $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, we define, point-wise, the Bellman Expectation operator $T_Q^\pi : \mathcal{Q} \rightarrow \mathcal{Q}$ as:

$$(T_Q^\pi f)(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|a, s) \sum_{a' \in \mathcal{A}} \pi(a'|s') f(s', a'), \quad \forall f \in \mathcal{Q} \quad (18)$$

This operator has a unique fixed point which corresponds to the action-value function q_π in our MDP $\mathcal{M}$. Same properties as T^π : γ -contraction and monotonicity.

Definition 9 (Bellman Optimality Operator for Q). Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, let $\mathcal{Q} \equiv \mathcal{Q}_{\mathcal{S}, \mathcal{A}}$ be the space of bounded real-valued functions over $\mathcal{S} \times \mathcal{A}$. We define the Bellman Optimality operator $T_Q^* : \mathcal{Q} \rightarrow \mathcal{Q}$ as:

$$(T_Q^* f)(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|a, s) \max_{a' \in \mathcal{A}} f(s', a'), \quad \forall f \in \mathcal{Q} \quad (19)$$

This operator has a unique fixed point which corresponds to the action-value function q^* in our MDP $\mathcal{M}$. Same properties as T^* : γ -contraction and monotonicity.

5 Approximate Dynamic Programming

5.1 Motivation

So far, we have assumed perfect knowledge of the MDP and perfect/exact representation of the value functions. Realistically, more often than not:

1. We won't know the underlying MDP.
 - Sampling/estimation error, as we don't have access to the true operators $T^\pi(T^*)$.
2. We won't be able to represent the value function exactly after each update.
 - Approximation error, as we approximate the true value functions within a (parametric) class (e.g. linear functions, neural nets, etc).

Objective: Under the above conditions, come up with a policy π that is (close to) optimal.

5.2 Approximate Value Iteration

Standard Value Iteration:

- Start with v_0 .
- Update values: $v_{k+1} = T^*v_k$.
- As $k \rightarrow \infty$, $v_k \rightarrow_{\|\cdot\|_\infty} v^*$.

Approximate Value Iteration:

- Start with v_0 .
- Update values: $v_{k+1} = \mathcal{A}T^*v_k$. ($v_{k+1} \approx T^*v_k$)

Question: As $k \rightarrow \infty$, $v_k \rightarrow v^*$?

Answer: In general, no.

5.2.1 Approximating the value function

Using a function approximator $v_\theta(s)$, with a parameter vector $\theta \in \mathbb{R}^m$.

- The estimated value function at iteration k is $v_k = v_{\theta_k}$.
- Use dynamic programming to compute $v_{\theta_{k+1}}$ from v_{θ_k} .
- $T^*v_k(s) = \max_a \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) | S_t = s]$.
- Fit θ_{k+1} s.t. $v_{\theta_{k+1}} \approx T^*v_k(s)$.
- For instance, with respect to a squared loss over the state-space:

$$\theta_{k+1} = \arg \min_{\theta_{k+1}} \sum_s (v_{\theta_{k+1}}(s) - T^*v_k(s))^2 \quad (20)$$

5.3 Example of divergence with dynamic programming

Tsitsiklis and Van Roy made an example where dynamic programming with linear function approximation can diverge. Consider the two state example below, where the rewards are all zero, there are no decisions, and there is a single parameter for estimating the value.

System Dynamics:

- State 1 (value θ) $\rightarrow$ State 2 (value 2θ).
- State 2 $\rightarrow$ State 2 (probability $1 - \epsilon$) or Terminal (probability ϵ).

Update Rule:

$$\theta_{k+1} = \operatorname{argmin}_{\theta} \sum_{s \in \mathcal{S}} (v_\theta(s) - \mathbb{E}[v_{\theta_k}(S_{t+1}) | S_t = s])^2 \quad (21)$$

$$= \operatorname{argmin}_{\theta} (\theta - \gamma 2\theta_k)^2 + (2\theta - \gamma(1 - \epsilon)2\theta_k)^2 \quad (22)$$

$$= \frac{2(3 - 2\epsilon)\gamma}{5} \theta_k \quad (23)$$

Divergence: What is $\lim_{k \rightarrow \infty} \theta_k$ when $\theta_0 = 1$, $\epsilon = \frac{1}{8}$ and $\gamma = 1$?

This is only a problem when we update the states, e.g., synchronously, without looking at the time an agent would spend in each state.

5.4 Hope for Approximate VI?

- As $k \rightarrow \infty$, $v_k \rightarrow_{\|\cdot\|_\infty} v^*$? **No.**
- Sample versions of these algorithms converge under mild conditions.
- Even for the function approximation case, the theoretical danger of divergence is rarely materialised in practice.
- There may be many value functions that can induce the optimal policy! (Example shown where different value functions induce same optimal policy).

6 Performance of a Greedy Policy

Theorem 4 (Value of greedy policy). *Consider an MDP. Let $q : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ be an arbitrary function and let π be the greedy policy associated with q , then:*

$$\|q^* - q^\pi\|_\infty \leq \frac{2\gamma}{1-\gamma} \|q^* - q\|_\infty \quad (24)$$

where q^* is the optimal value function associated with this MDP.

Proof. Statement: $\|q^* - q^\pi\|_\infty \leq \frac{2\gamma}{1-\gamma} \|q^* - q\|_\infty$.

$$\|q^* - q^\pi\|_\infty = \|q^* - T^\pi q + T^\pi q - q^\pi\|_\infty \quad (15)$$

$$\leq \|q^* - T^\pi q\|_\infty + \|T^\pi q - q^\pi\|_\infty \quad (16)$$

$$= \|T^* q^* - T^* q\|_\infty + \|T^\pi q - T^\pi q^\pi\|_\infty \quad (17)$$

$$\leq \gamma \|q^* - q\|_\infty + \gamma \|q - q^\pi\|_\infty \quad (18)$$

$$\leq \gamma \|q^* - q\|_\infty + \gamma (\|q - q^*\|_\infty + \|q^* - q^\pi\|_\infty) \quad (19)$$

Re-arranging:

$$(1 - \gamma) \|q^* - q^\pi\|_\infty \leq 2\gamma \|q^* - q\|_\infty \quad (25)$$

□

Observations:

- Small values of γ obtain a better (smaller) upper bound on the potential loss of performance.
- In particular, what happens for $\gamma = 0$?
- What if $q = q^*$? What does this bound imply in that case?

7 Approximate Policy Iteration

Standard Policy Iteration:

- Start with π_0 .
- Iterate:
 - Policy Evaluation: $q_i = q_{\pi_i}$
 - Greedy Improvement: $\pi_{i+1} = \arg \max_a q_{\pi_i}(s, a)$
- As $i \rightarrow \infty$, $q_i \rightarrow_{\|\cdot\|_\infty} q^*$. Thus $\pi_i \rightarrow \pi^*$.

Approximate Policy Iteration:

- Start with π_0 .
- Iterate:
 - Policy Evaluation: $q_i = \mathcal{A}q_{\pi_i}$ ($q_i \approx q_{\pi_i}$)
 - Greedy Improvement: $\pi_{i+1} = \arg \max_a q_i(s, a)$

Question 1: As $i \rightarrow \infty$, does $q_i \rightarrow_{\|\cdot\|_\infty} q^*$?

Answer: In general, no.

Question 2: Or does π_i converge to the optimal policy?

Answer: In general, no.

Hopeless? In some cases, no, depending on the nature of $\mathcal{A}$. (More: Next lectures)